

● MEDIUM

● PROBLEM-SOLVING AND DATA ANALYSIS

● ANSWER KEY

Problem-Solving and Data Analysis

01

10 answers · Rationale-rich · Teach from doc

Q1**450**

The correct answer is 450. Let x represent the number that 432 is 96% of. This can be written as $\frac{96}{100}x = 432$, or $0.96x = 432$. Dividing both sides of this equation by 0.96 yields $x = 450$. Therefore, 432 is 96% of 450.

Q2**D**

D. $\frac{12}{16}$

Choice D is correct. The table shows that there are a total of 16 kittens that have a chocolate-colored coat. Of the 16 with a chocolate-colored coat, 12 have deep blue eyes. Therefore, the fraction of chocolate-colored kittens with deep blue eyes is simply the ratio of those two numbers, or $\frac{12}{16}$.

Choice A is incorrect; this is the fraction of all chocolate-colored kittens. Choice B is incorrect; this is the fraction of kittens with deep blue eyes that have a chocolate-colored coat. Choice C is incorrect; this is the fraction of cream-tortoiseshell-colored kittens with deep blue eyes.

Q3**B****B. -1.1**

Choice B is correct. A line in the xy -plane that passes through points x_1, y_1 and x_2, y_2 has a slope of $\frac{y_2 - y_1}{x_2 - x_1}$. The line of best fit shown passes approximately through the points 0, 14 and 13, 0. It follows that the slope of this line of best fit is approximately $\frac{0 - 14}{13 - 0}$, or $-\frac{14}{13}$. Of the given choices, -1.1 is closest to $-\frac{14}{13}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q4

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Answer not provided in source data — see rationale below.

The correct answer is $\frac{1}{5}$. The mean of the list of numbers is found by dividing the sum of the numbers by the number of values in the list. Since there are 5 numbers in the list, the mean is $\frac{1}{5}$ of the sum of the numbers. Note that $\frac{1}{5}$ and .2 are examples of ways to enter a correct answer.

Q5**17**

The correct answer is 17. If one value is multiplied by a number, then the other value must be multiplied by the same number in order to maintain the same ratio. It's given that j is multiplied by 17. Therefore, in order to maintain the same ratio, k must also be multiplied by 17.

Q6

B

B. It is plausible that between 8.38% and 15.62% of the socks are defective.

Choice B is correct. It's given that, based on the sample, an estimate of 12% of all socks produced by the company in a certain week are defective, with an associated margin of error of 3.62%. This estimate, plus or minus the margin of error, gives an interval of plausible values for the actual percent of all socks produced by the company that week that are defective.

Subtracting 3.62% from 12% yields 8.38%. Adding 3.62% to 12% yields 15.62%. Therefore, it is plausible that between 8.38% and 15.62% of all socks produced by the company are defective.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect. 12% is the estimated percent of defective socks based on the sample. However, since the margin of error for this estimate is known, the most appropriate conclusion is not that the percent of defective socks is exactly 12% but instead that it lies in an interval of plausible percents.

Choice D is incorrect and may result from conceptual errors.

Q7

A

A. Neither

Choice A is correct. The purpose of surveying a random sample of residents is to approximate the percent of the town residents that are satisfied with the concession stand. The sample doesn't necessarily get the same result as surveying every resident of the town, nor would another sample necessarily have identical results. Therefore, although it's possible that either statement I or statement II could prove true by surveying every resident of the town, these statements cannot be proven true solely based on the results of the sample.

Choice B is incorrect because surveying a sample of the town residents may not have the same result as surveying all the town residents. Choices C and D are incorrect because surveying a different sample of residents could yield different results.

Q8**324**

The correct answer is 324. It's given that 1,250 people attended the first information session, and that 72% of the people who attended the first information session attended the second information session. Therefore, the number of people who attended the first and second information sessions can be found by calculating 72% of 1,250, which is equal to $1,250 \frac{72}{100}$, or 900. It's also given that 36% of the people who attended the first and second information sessions attended the third information session. Since 900 people attended the first and second information sessions, the number of people who attended the first, second, and third information sessions can be found by calculating 36% of 900, which is equal to $900 \frac{36}{100}$, or 324. Therefore, 324 people attended all three information sessions.

Q9**A****A. Chinstrap**

Choice A is correct. It is given that there are 180 female penguins in the exhibit. Therefore, $\frac{1}{3}$ of the female penguins is $\frac{1}{3} \times 180 = 60$ penguins. According to the table, there are 59 female chinstrap penguins, 27 female emperor penguins, 54 female gentoo penguins, and 40 female macaroni penguins. So the female chinstrap penguin population is the closest to 60, or $\frac{1}{3}$ of the total female population in the exhibit.

Choices B, C, and D are incorrect and may result from reading data from the table incorrectly. Since the total female penguin population is 180, $\frac{1}{3}$ of the total female penguin population is 60. The numbers of female emperor (27), female gentoo (54), and female macaroni (40) penguins are not as close to 60 as the number of female chinstrap penguins (59).

Q10

A

A. $y = 3x + 0.8$

Choice A is correct. The data show a strong linear relationship between x and y . The line of best fit for a set of data is a linear equation that minimizes the distances from the data points to the line. An equation for the line of best fit can be written in slope-intercept form, $y = mx + b$, where m is the slope of the graph of the line and b is the y -coordinate of the y -intercept of the graph. Since, for the data shown, the y -values increase as the x -values increase, the slope of a line of best fit must be positive. The data shown lie almost in a line, so the slope can be roughly estimated using the formula for slope, $m = \frac{y_2 - y_1}{x_2 - x_1}$. The leftmost and rightmost data points have coordinates of about $(1, 4)$ and $(8, 26)$, so the slope is approximately $\frac{26 - 4}{8 - 1} = \frac{22}{7}$, which is a little greater than 3. Extension of the line to the left would intersect the y -axis at about $(0, 1)$. Only choice A represents a line with a slope close to 3 and a y -intercept close to $(0, 1)$.

Choice B is incorrect and may result from switching the slope and y -intercept. The line with a y -intercept of $(0, 3)$ and a slope of 0.8 is farther from the data points than the line with a slope of 3 and a y -intercept of $(0, 0.8)$. Choices C and D are incorrect. They represent lines with negative slopes, not positive slopes.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

Answers recovered from rationale for Q4 — the source CB data omitted a structured key for these items, so the answer slot shows —. The worked solution in the rationale below each entry contains the correct value.

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